# _eda_price

March 30, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
customers = aggregate_customer(df_clean)
```

```
Quote-level columns created:
  - out_of_pocket: 34,991 values
  - subsidy_issue: 9,076 issues (24.5%)

 Customer dataset created: 25,716 customers
 Total columns: 52
 Key columns now available:
   - decision_days: from customer_duration_days
   - price_cv: from std_out_of_pocket / avg_out_of_pocket
   - quote_count: alias for total_quotes
   - main_brand: most common brand per customer
   - price_range: 1142 avg
   - price_volatility: 0.26 avg
subsidy_issue_type
No Issues            18660
Both                  2480
CEE Only              2361
MaPrimeRénov Only     2215
Name: count, dtype: int64

 Saved to 'customer_master_data.csv'
```

```python
customers = pd.read_csv('customer_master_data.csv')
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)
```

```
visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.363130           18660           6776.0
True                      0.476616            7056           3363.0

Chi-square p-value: 6.718504e-62
Statistically significant: YES
```

```
Conversion without issues: 36.31%
Conversion with issues: 47.66%
Difference: 11.3 percentage points
Customer-level finding: +11.3 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.40
  With issues: 1.55
  Ratio: 1.11x

avg_quote_amount:
  No issues: 6137.73
  With issues: 9895.80
  Ratio: 1.61x

max_quote_amount:
  No issues: 6557.78
  With issues: 10828.77
  Ratio: 1.65x

customer_duration_days:
  No issues: 19.20
  With issues: 26.05
  Ratio: 1.36x

price_range:
  No issues: 899.87
  With issues: 1770.19
  Ratio: 1.97x

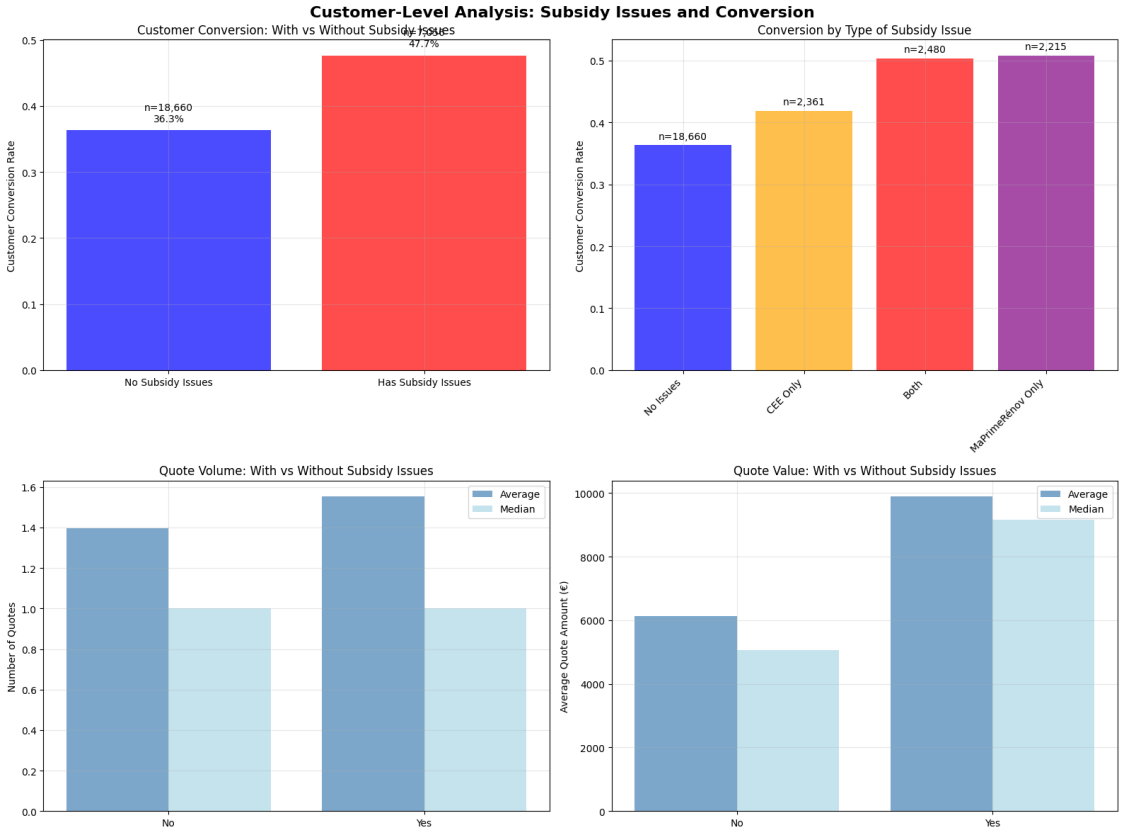
multiple_quotes:
  No issues: 0.28
  With issues: 0.36
  Ratio: 1.28x

Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.717  0.636
Three+ quotes      0.070  0.117
Two quotes         0.214  0.247
```

```
Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
Long (>30 days)     0.073  0.107
Medium (8-30 days)  0.032  0.046
Quick (2-7 days)    0.041  0.066
Same day decision   0.854  0.781

Product family by subsidy issue status:
had_subsidy_issue                False  True
main_product_family
Accessoire de pose               0.004  0.000
Appareil hybride                 0.002  0.018
Autres                           0.013  0.005
Chaudière                        0.364  0.048
Climatisation                    0.259  0.072
ECS : Chauffe-eau ou adoucisseur 0.065  0.025
Emetteur de chauffage  ou chappe 0.012  0.011
Photovoltaïque                   0.029  0.015
Plomberie Sanitaire              0.001  0.000
Pompe à chaleur                  0.068  0.432
Poêle                            0.171  0.374
Produit VMC                      0.003  0.000
Régulation ou électricité        0.000  0.000
Unknown                          0.009  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.508352   2215
Both                0.503226   2480
CEE Only            0.418890   2361
No Issues           0.363130  18660
```




```
================================================================================
Policy Suspension Impact
================================================================================

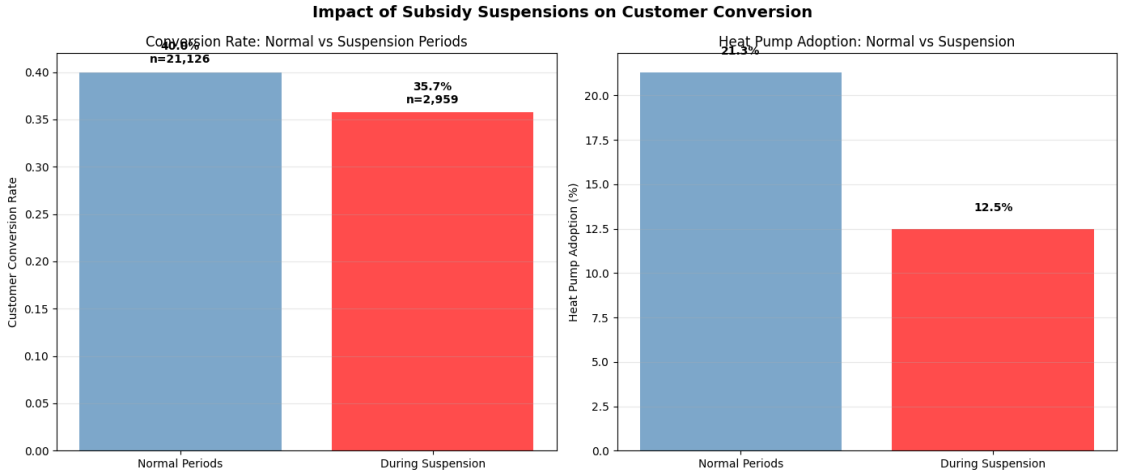
Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
Unique values in during_suspension: [False  True]

Conversion during subsidy suspensions:
                      mean  count
Normal Periods     0.399886  21126
During Suspension  0.357215   2959

Suspension impact p-value: 0.0000
Statistically significant: YES

Product mix during suspensions:
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
during_suspension
```

```
False                          19.4     28.4        18.5     9.9    23.7
True                           37.9     27.4        10.2     7.5    17.1
```



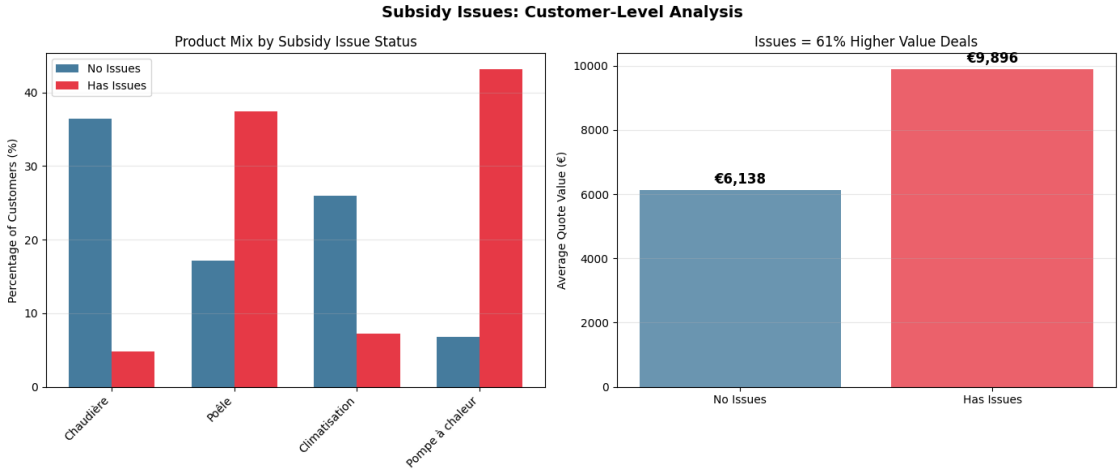

```
Unique values in main_product_family:
main_product_family
Chaudière                           7127
Poêle                               5838
Climatisation                       5342
Pompe à chaleur                     4322
ECS : Chauffe-eau ou adoucisseur    1386
Photovoltaïque                       645
Emetteur de chauffage  ou chappe     310
Autres                               279
Unknown                              168
Appareil hybride                     157
Accessoire de pose                    70
Produit VMC                           58
Plomberie Sanitaire                   13
Régulation ou électricité              1
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                           7127
Poêle                               5838
Climatisation                       5342
Pompe à chaleur                     4322
ECS : Chauffe-eau ou adoucisseur    1386
Photovoltaïque                       645
Emetteur de chauffage  ou chappe     310
Autres                               279
```

```
Unknown                                168
Appareil hybride                       157
Name: count, dtype: int64

Using these products: ['Chaudière', 'Poêle', 'Climatisation', 'Pompe à chaleur']

No Issue %: [np.float64(36.37727759914255), np.float64(17.138263665594856),
np.float64(25.905680600214364), np.float64(6.843515541264737)]
Has Issue %: [np.float64(4.804421768707483), np.float64(37.41496598639456),
np.float64(7.199546485260772), np.float64(43.154761904761905)]
```

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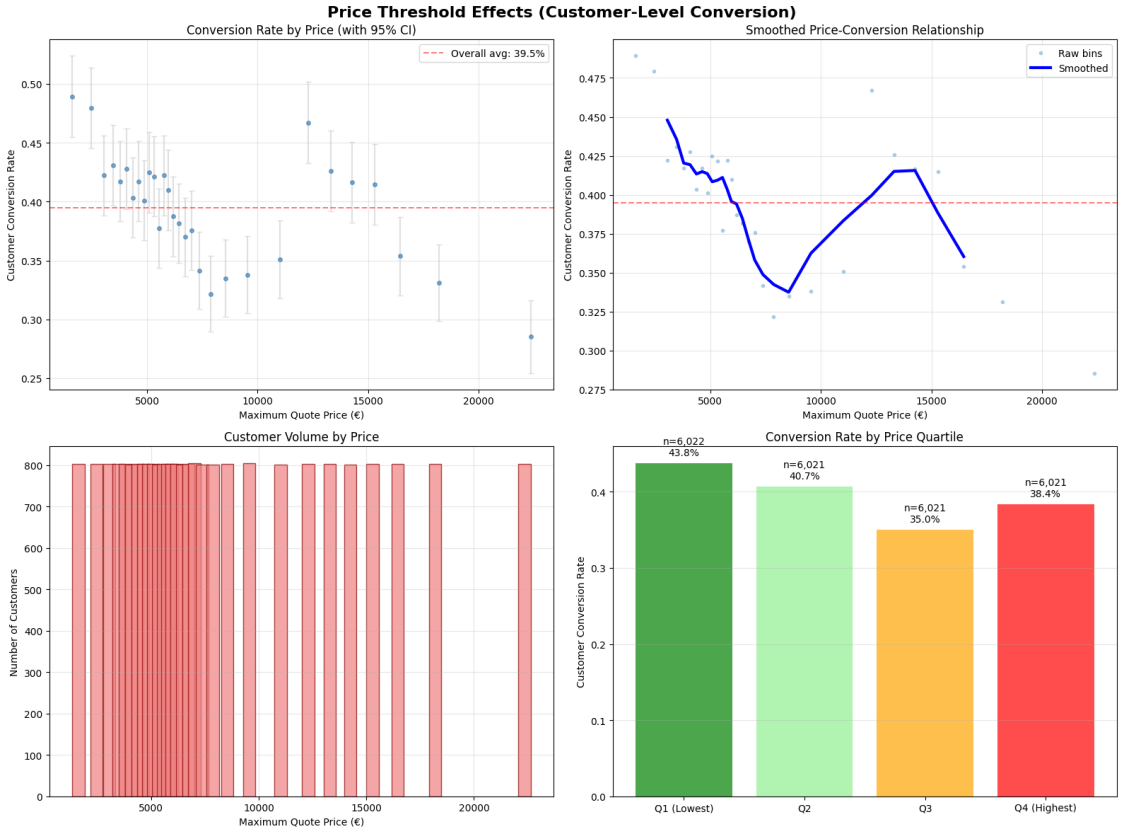
================================================================================
Threshold effect
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```

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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 43.8% (n=6,022)
  Q2: 40.7% (n=6,021)
  Q3: 35.0% (n=6,021)
  Q4 (Highest): 38.4% (n=6,021)

Difference between lowest and highest quartile: 5.4 percentage points

Sweet spots (above-average conversion):
  €1,622: 48.9% (n=803)
  €2,460: 47.9% (n=803)
  €3,057: 42.2% (n=803)
  €3,475: 43.1% (n=803)
  €3,794: 41.7% (n=803)
  €4,081: 42.8% (n=802)
  €4,370: 40.4% (n=803)
```

```
  €4,632: 41.7% (n=803)
  €4,861: 40.1% (n=803)
  €5,084: 42.5% (n=803)
  €5,326: 42.1% (n=802)
  €5,759: 42.2% (n=803)
  €5,954: 41.0% (n=803)
  €12,299: 46.7% (n=803)
  €13,308: 42.6% (n=803)
  €14,256: 41.6% (n=802)
  €15,286: 41.5% (n=803)

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
price_quartile
Q1_Low                   25.5    39.3        0.7   24.6   10.0
Q2_MidLow                16.4    34.5        0.5    3.1   45.6
Q3_MidHigh               27.7    29.7        4.7    5.3   32.7
Q4_High                  16.8     8.1       64.9    8.7    1.5

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=4,662):
  Q1_Low: 50.0%
  Q2_MidLow: 58.6%
  Q3_MidHigh: 57.8%
  Q4_High: 40.5%

Boiler (n=7,110):
  Q1_Low: 44.6%
  Q2_MidLow: 39.8%
  Q3_MidHigh: 38.6%
  Q4_High: 35.9%

AC (n=5,451):
  Q1_Low: 37.3%
  Q2_MidLow: 30.6%
  Q3_MidHigh: 28.9%
  Q4_High: 31.5%
```

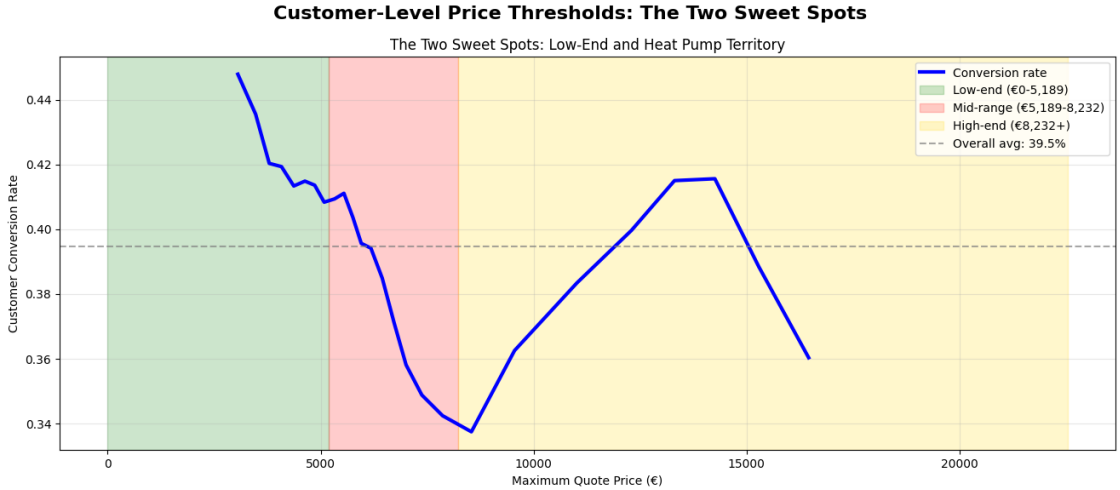
```
Stove (n=5,831):
  Q1_Low: 49.8%
  Q2_MidLow: 44.0%
  Q3_MidHigh: 33.8%
  Q4_High: 39.1%

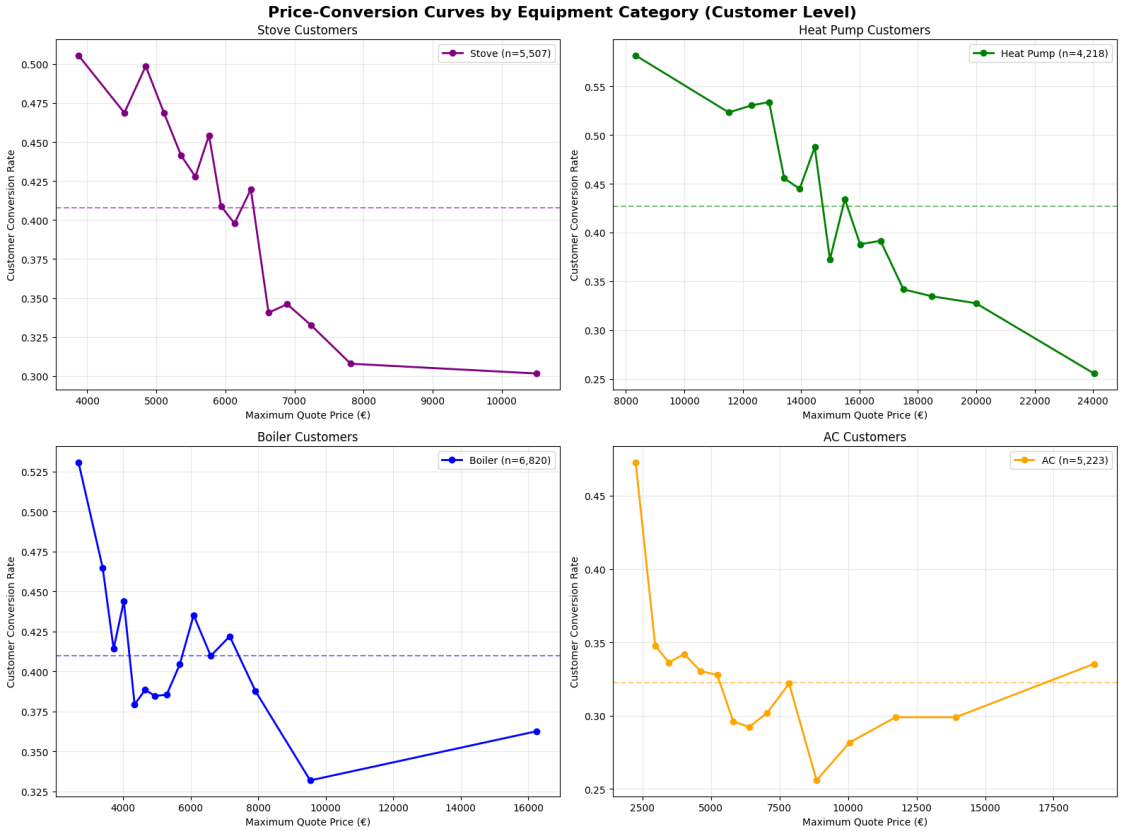
================================================================================
HEAT PUMP CUSTOMERS (n=5,357)
================================================================================
Average price: €16,032
Conversion rate: 41.6%
Subsidy issue rate: 70.1%

Heat pump adoption by quartile:
  Q1_Low: 0.9%
  Q2_MidLow: 0.6%
  Q3_MidHigh: 5.6%
  Q4_High: 74.8%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```



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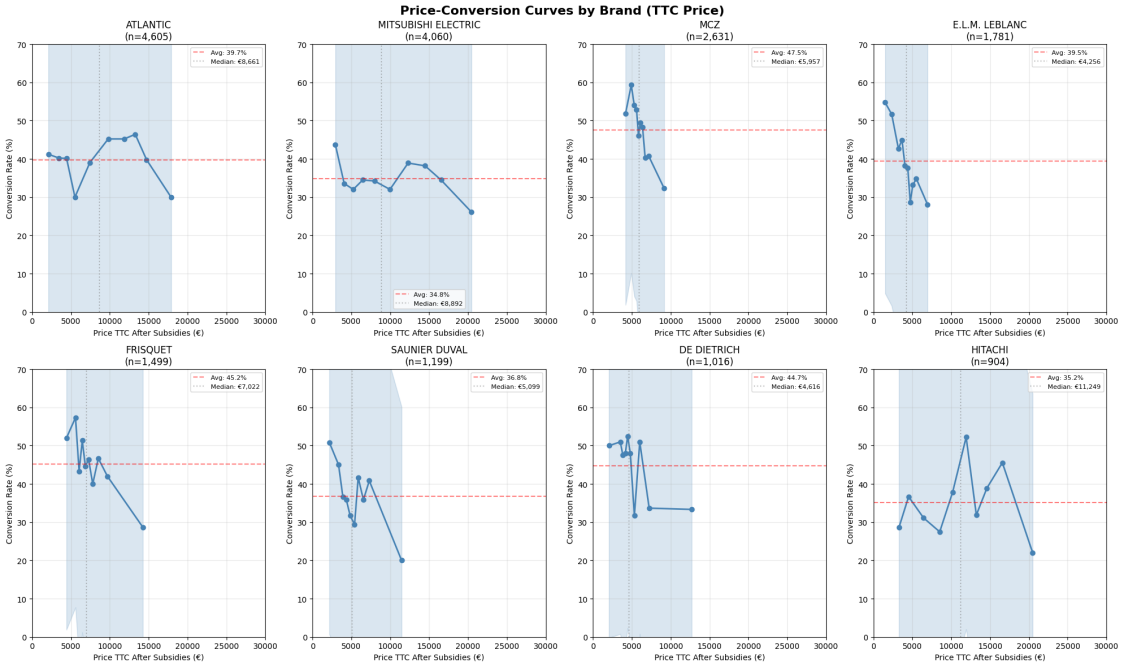
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'E.L.M. LEBLANC',
'FRISQUET', 'SAUNIER DUVAL', 'DE DIETRICH', 'HITACHI']
```

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Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
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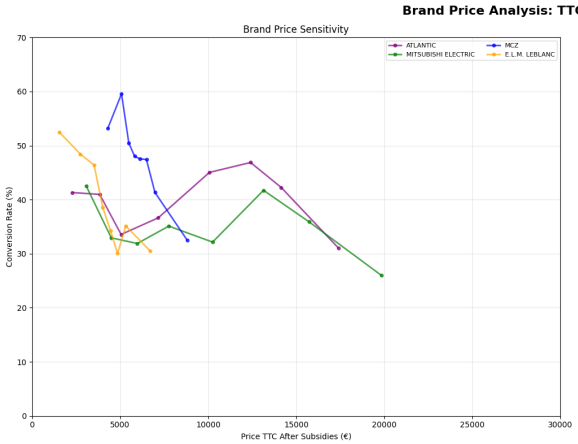



**Brand Performance Summary**

| Brand | Customers | Avg Conv | Avg Price | Best Price Range | Best Conv |
|---|---|---|---|---|---|
| ATLANTIC | 4,605 | 39.7% | €9,043 | €10,915-13,943 | 45.8% |
| MITSUBISHI ELECTRIC | 4,060 | 34.8% | €10,035 | €1,368-4,644 | 38.7% |
| MCZ | 2,631 | 47.5% | €6,136 | €1,800-5,145 | 55.6% |
| E.L.M. LEBLANC | 1,781 | 39.5% | €4,146 | €1,328-2,833 | 53.2% |
| FRISQUET | 1,499 | 45.2% | €7,685 | €2,391-5,829 | 54.7% |
| SAUNIER DUVAL | 1,199 | 36.8% | €5,515 | €1,332-3,677 | 47.9% |
| DE DIETRICH | 1,016 | 44.7% | €5,408 | €1,336-3,680 | 50.5% |
| HITACHI | 904 | 35.2% | €10,983 | €9,201-12,631 | 45.0% |

```
================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 24,085 customers
```

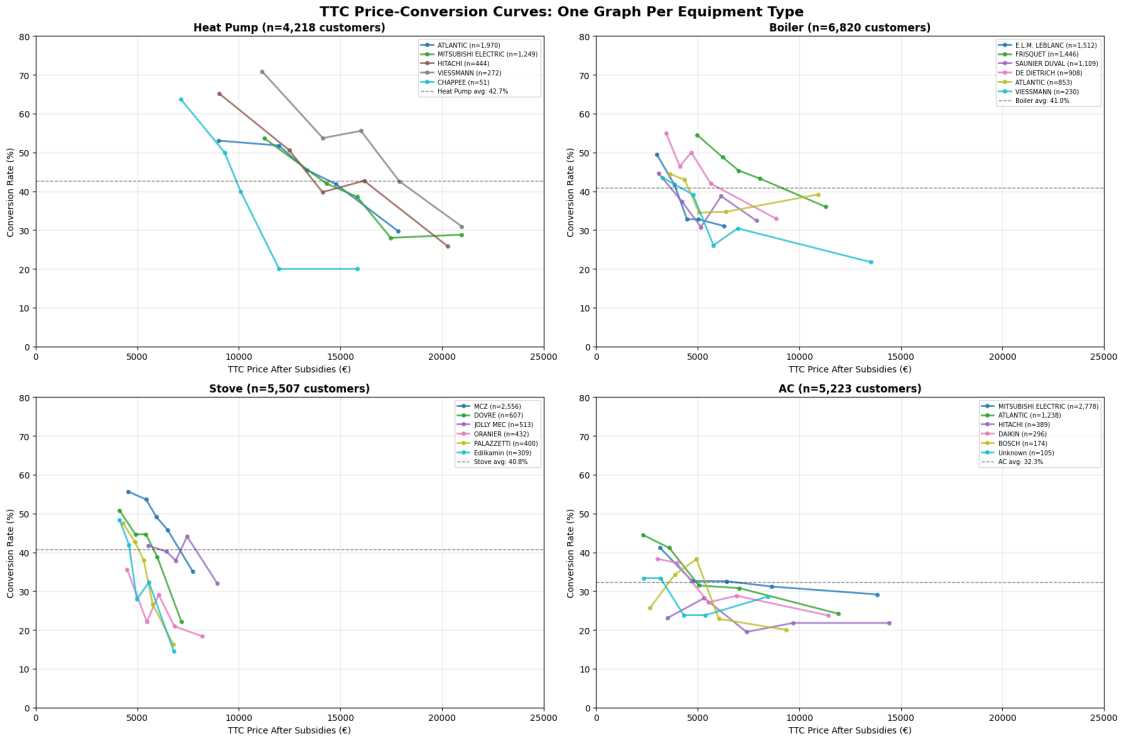
```
Price range: €1,332 to €28,168
Median price: €6,309

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'HITACHI',
'VIESSMANN', 'CHAPPEE']

Boiler - Top brands: ['E.L.M. LEBLANC', 'FRISQUET', 'SAUNIER DUVAL', 'DE
DIETRICH', 'ATLANTIC', 'VIESSMANN']

Stove - Top brands: ['MCZ', 'DOVRE', 'JOLLY MEC', 'ORANIER', 'PALAZZETTI',
'Edilkamin']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC', 'HITACHI', 'DAIKIN',
'BOSCH', 'Unknown']
```

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================================================================================
Seasonal Effects on Customer Conversion
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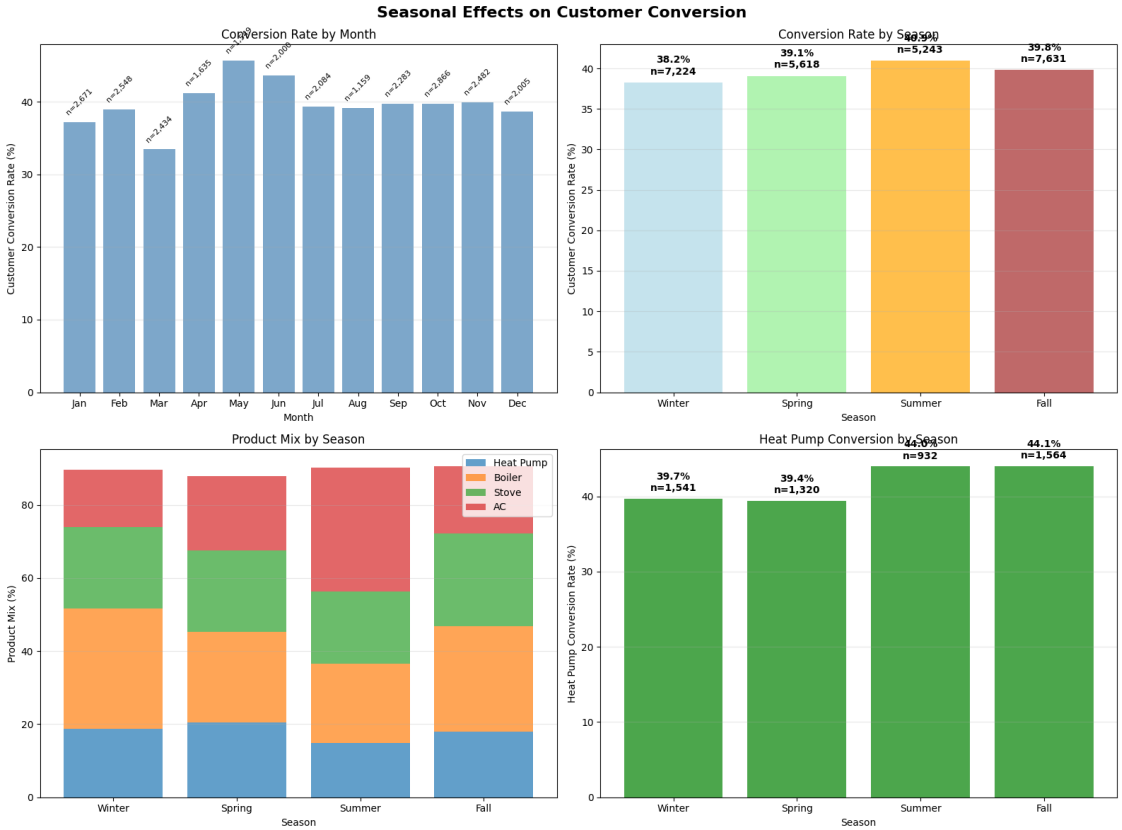


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================================================================================
SEASONAL STATISTICS
================================================================================

Conversion by season:
  Winter: 38.2% (n=7,224)
  Spring: 39.1% (n=5,618)
  Summer: 40.9% (n=5,243)
  Fall: 39.8% (n=7,631)

Seasonality chi-square p-value: 0.0193
Statistically significant: YES

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```

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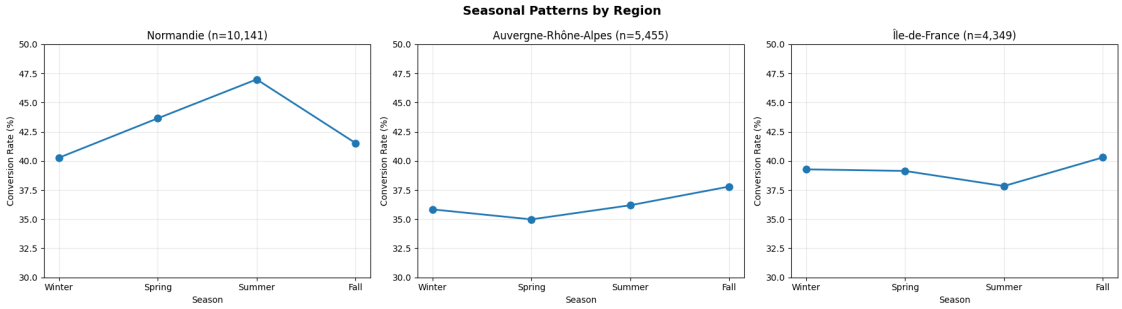
Top regions: ['Normandie', 'Auvergne-Rhône-Alpes', 'Île-de-France']
```

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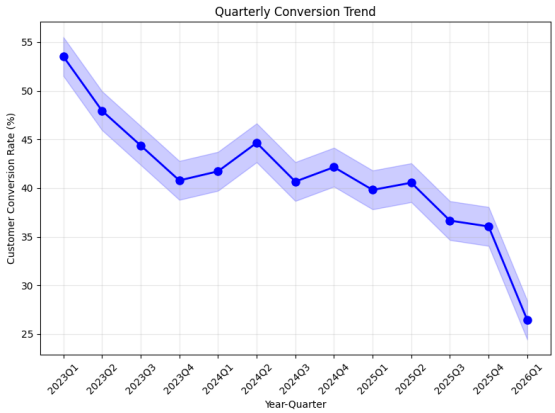
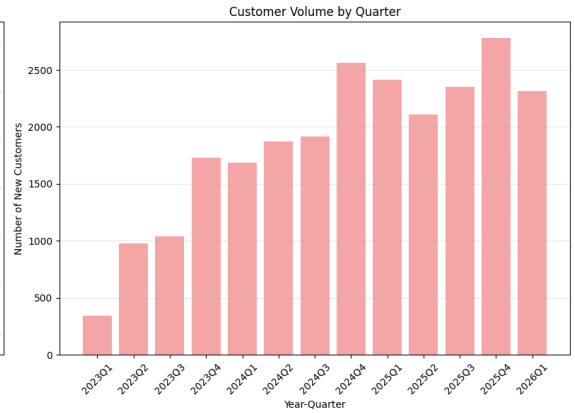
================================================================================
Year-over-year trends
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

Quarterly conversion rates:
                  mean  count
year_quarter
2023Q1        0.535294    340
2023Q2        0.479550    978
2023Q3        0.443804   1041
2023Q4        0.407857   1731
2024Q1        0.417211   1685
2024Q2        0.446524   1870
2024Q3        0.406691   1913
2024Q4        0.421484   2560
2025Q1        0.398094   2414
2025Q2        0.405508   2106
2025Q3        0.366539   2349
2025Q4        0.360532   2782
2026Q1        0.264249   2316
```

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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

Conversion during subsidy suspensions:
                       mean  count
Normal Periods     0.399886  21126
During Suspension  0.357215   2959

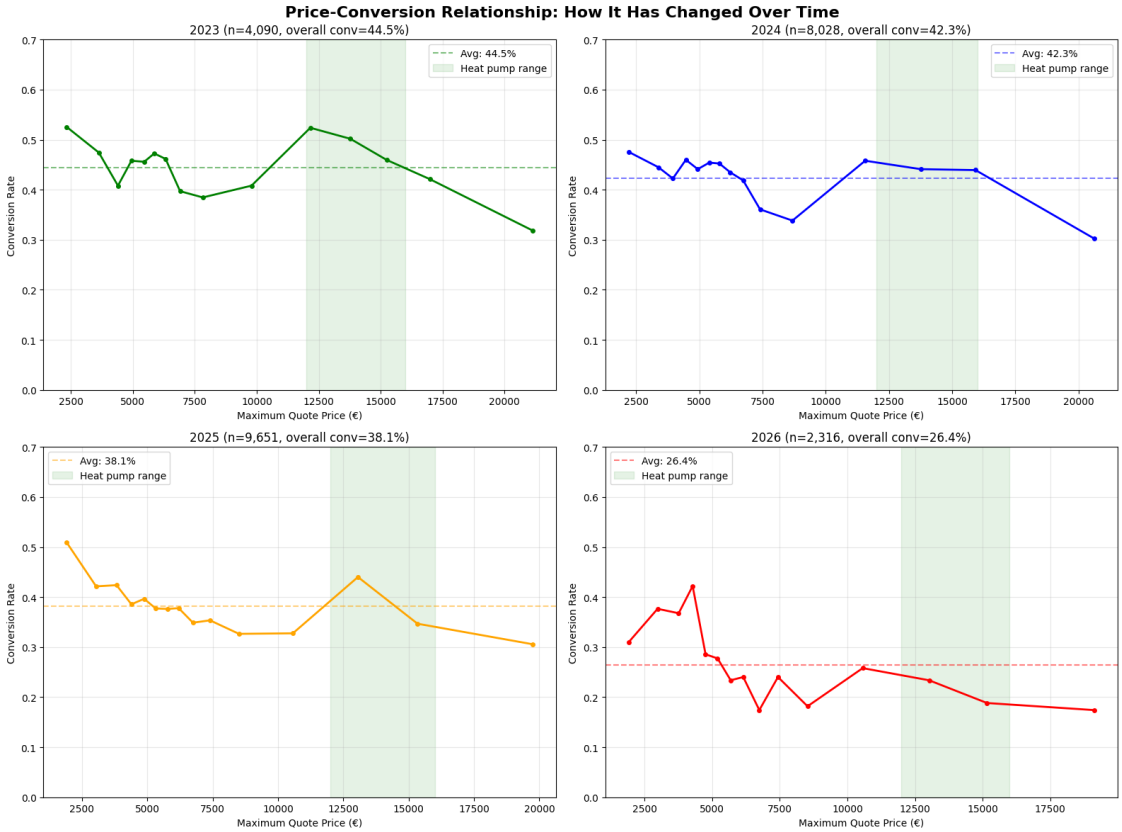
Suspension impact p-value: 0.0000
Statistically significant: YES

Best month: 5 (45.4%, n=1483)
Worst month: 3 (33.4%, n=2195)

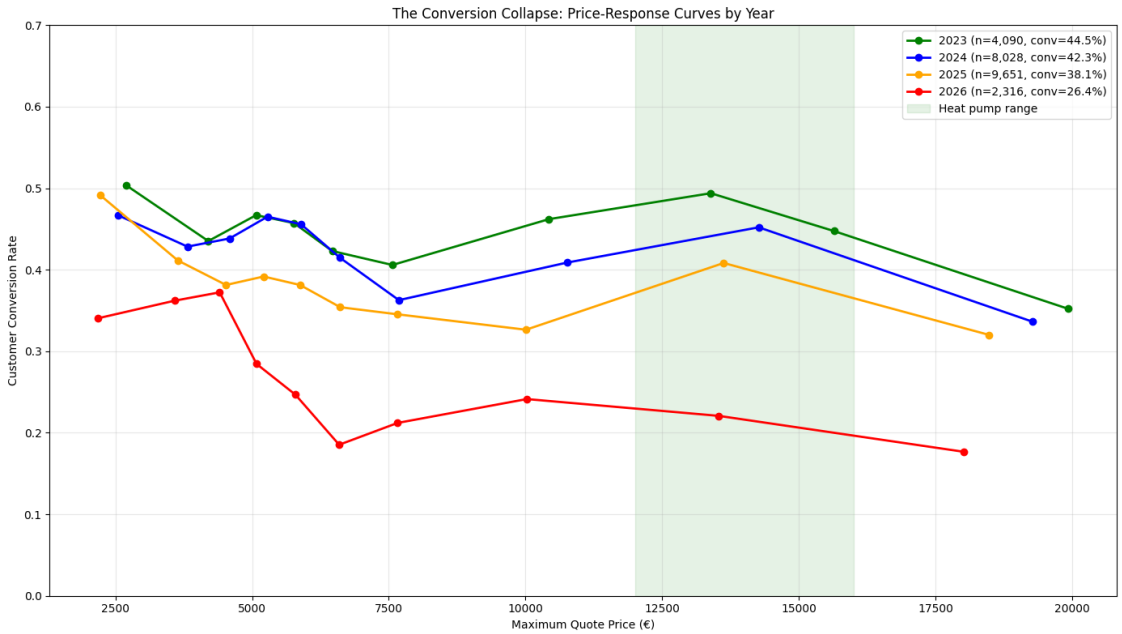
Best season: Summer (41.2%)
Worst season: Winter (38.1%)

Heat pump conversion by season:
  Winter: 40.9%
  Spring: 40.4%
  Summer: 45.7%
  Fall: 44.7%

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```




```
Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```




```
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

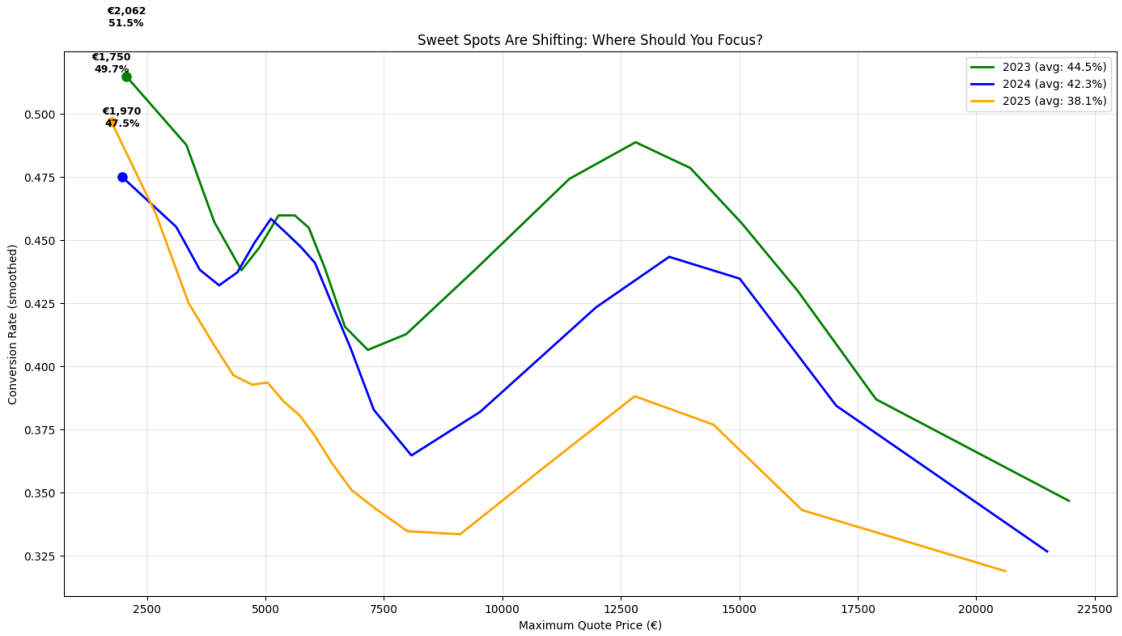
Sweet spots by year (price points with conversion > avg + 3%):

2023 (overall avg: 44.5%):
  €2,062: 54.1% (n=205)
  €3,920: 48.1% (n=210)
  €11,413: 50.7% (n=205)
  €12,812: 49.0% (n=204)
  €13,965: 49.8% (n=205)

2024 (overall avg: 42.3%):
  €1,970: 49.5% (n=402)
  €5,115: 49.4% (n=401)
  €6,043: 47.1% (n=401)
  €11,976: 48.6% (n=401)
  €15,010: 49.0% (n=402)

2025 (overall avg: 38.1%):
  €1,750: 52.4% (n=483)
```

```
€2,684: 46.0% (n=483)
€3,902: 43.1% (n=483)
€5,043: 42.1% (n=482)
€12,793: 43.6% (n=482)
```

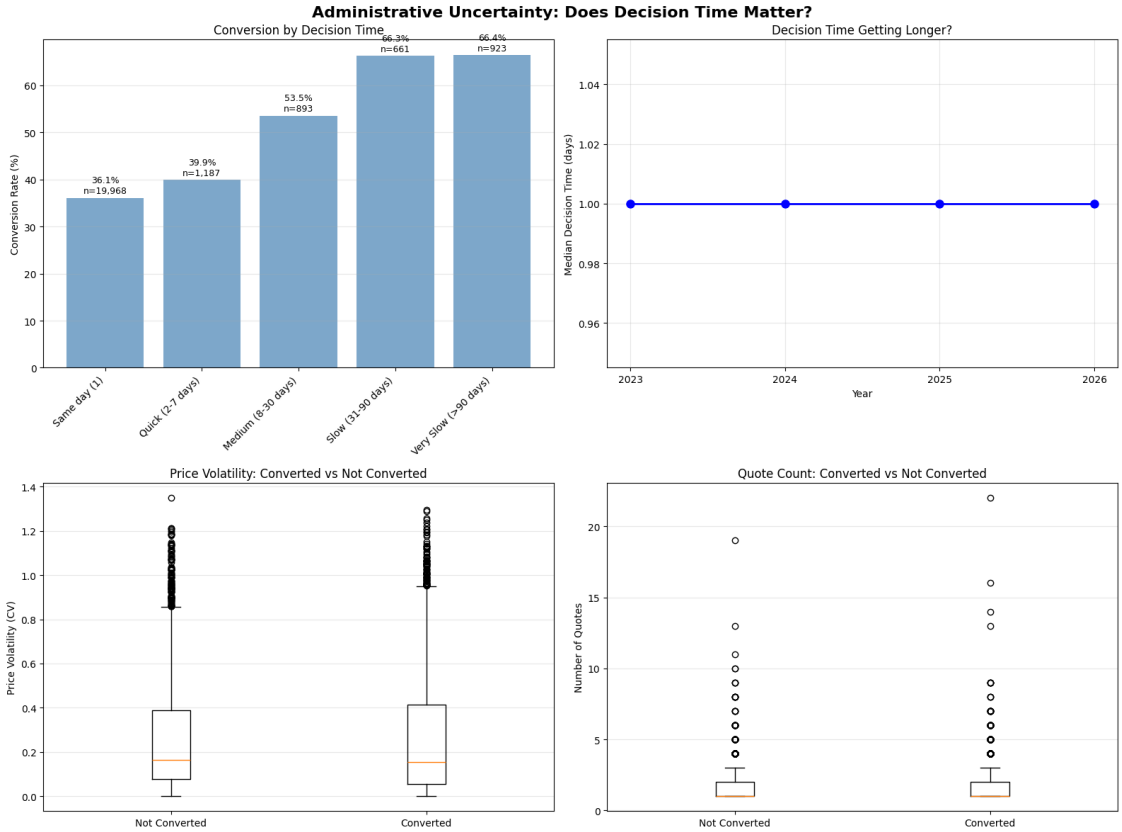



```
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

Decision time distribution:
count    24085.000000
mean        21.708034
std         91.365934
min          1.000000
25%          1.000000
50%          1.000000
75%          1.000000
max       1085.000000
Name: decision_days, dtype: float64

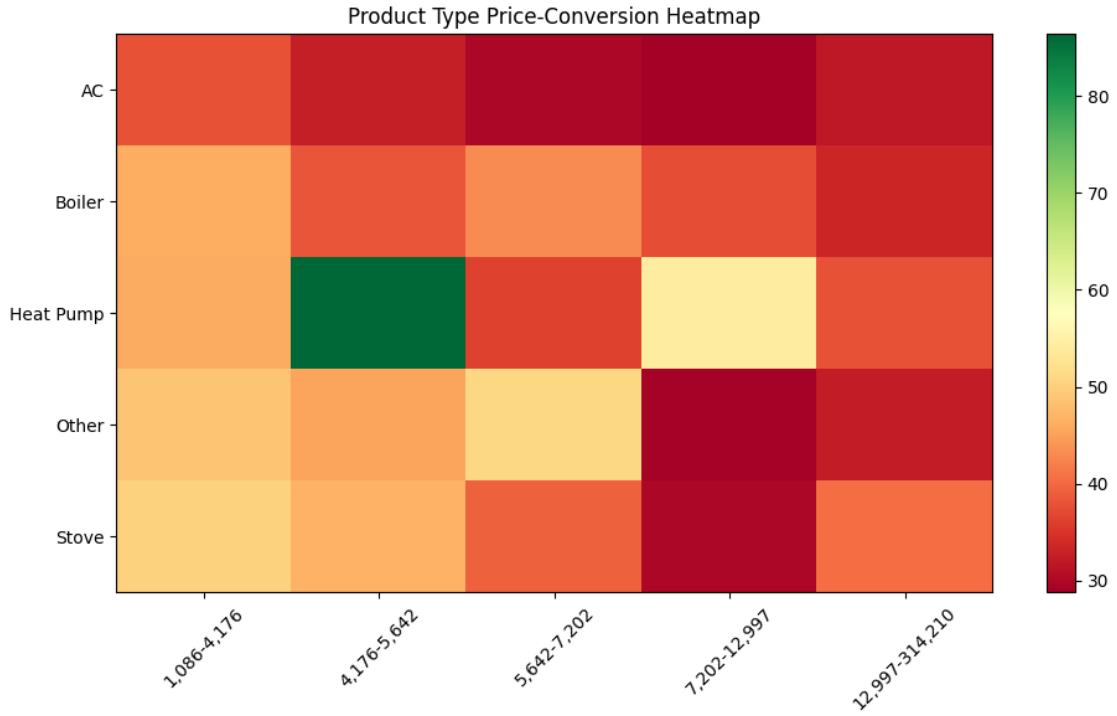
Customers with 1-day decision: 19,968 (82.9%)
```




```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1086.579, 4176.528]  (4176.528, 5642.004]  \
main_equipment_category
AC                                37.839879             32.461356
Boiler                            46.158068             38.014888
Heat Pump                         45.945946             86.363636
Other                             48.797251             45.398773
Stove                             50.180505             46.744931

price_bin_simple         (5642.004, 7202.244]  (7202.244, 12997.88]  \
main_equipment_category
AC                                29.751131             28.793774
Boiler                            43.175487             37.415541
Heat Pump                         36.363636             54.205607
Other                             50.909091             29.216867
Stove                             39.327243             29.865772
```

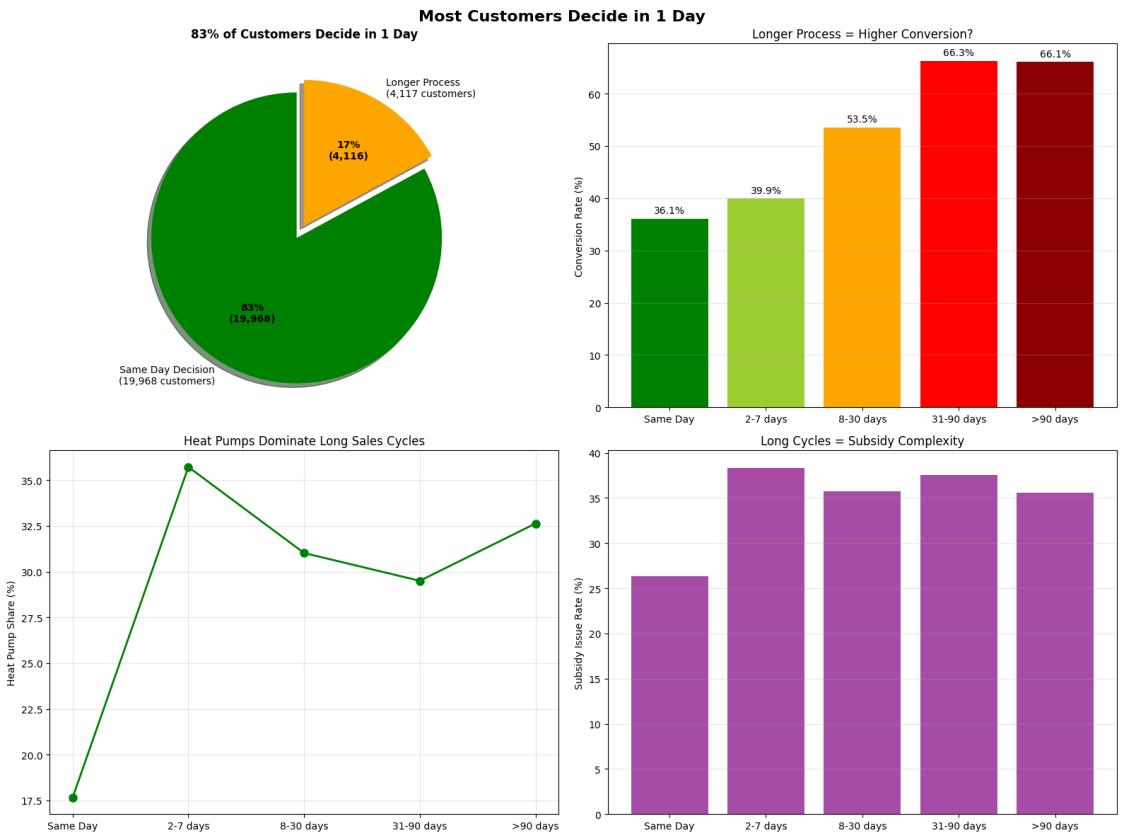
```
price_bin_simple         (12997.88, 314210.61]
main_equipment_category
AC                                   31.748252
Boiler                               33.333333
Heat Pump                            37.988488
Other                                32.199546
Stove                                40.322581
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          7127
Poêle                              5838
Climatisation                      5342
Pompe à chaleur                    4322
ECS : Chauffe-eau ou adoucisseur   1386
Photovoltaïque                      645
Emetteur de chauffage  ou chappe    310
Autres                              279
Unknown                             168
Appareil hybride                    157
Accessoire de pose                   70
```

```
Produit VMC                          58
Plomberie Sanitaire                  13
Régulation ou électricité             1
Name: count, dtype: int64

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309
```

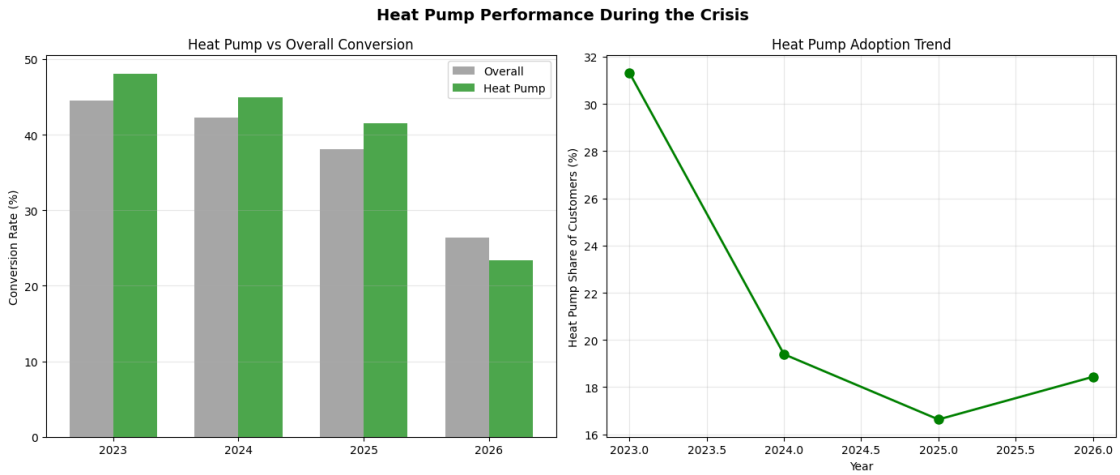



```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
2023              0.481            1281  15331.030               0.732
```

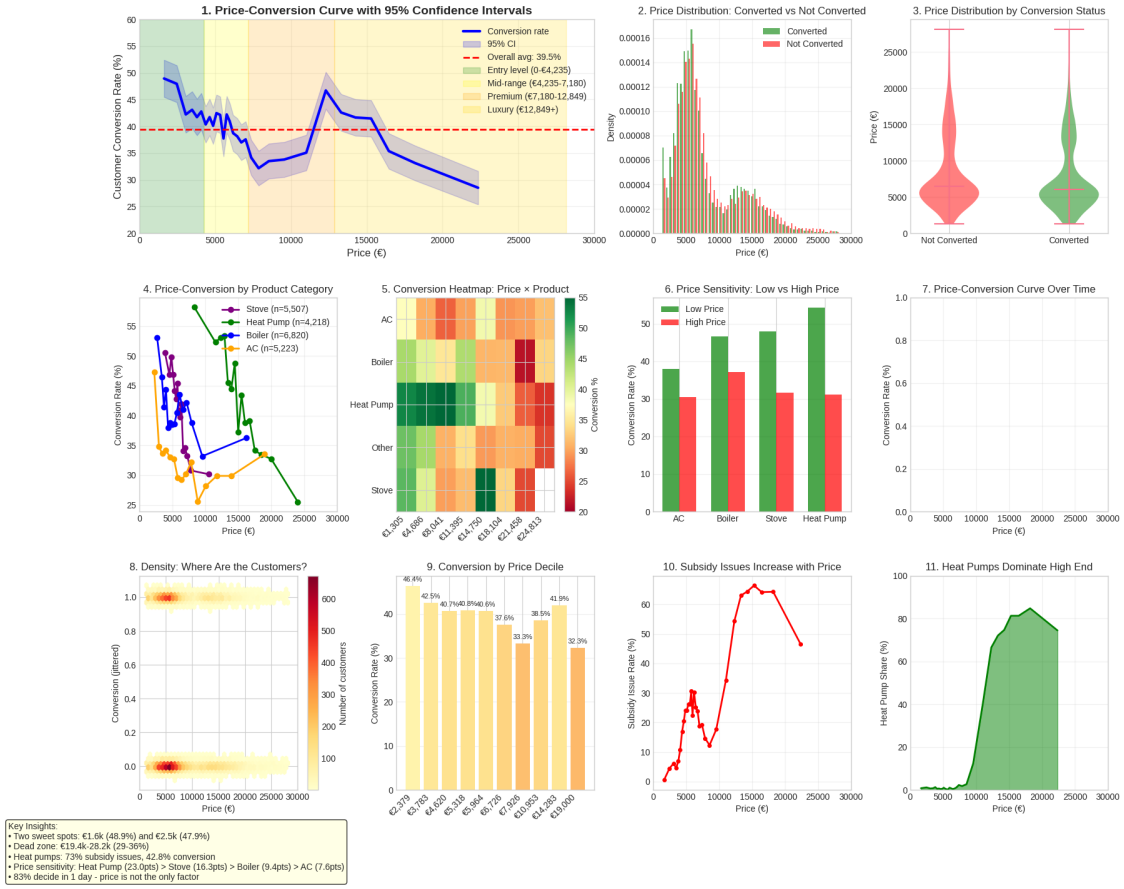
| | | | | |
|---|---|---|---|---|
| 2024 | 0.450 | 1557 | 15538.494 | 0.680 |
| 2025 | 0.415 | 1605 | 15290.053 | 0.753 |
| 2026 | 0.234 | 427 | 15032.380 | 0.773 |



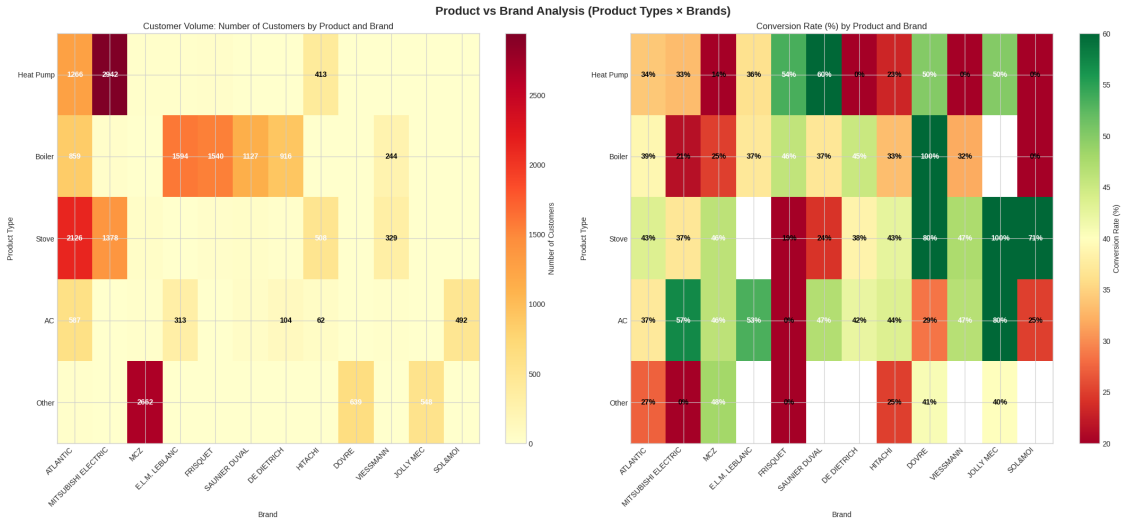

```
Analyzing 24,085 customers
Price range: €1,332 to €28,168
Median price: €6,309

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €1.6k (48.9%) and €2.5k (47.9%)
Dead zone: €19.4k-28.2k (29-36%)
Heat pumps: 73% subsidy issues, 42.8% conversion
Price sensitivity: Heat Pump (23.0pts) > Stove (16.3pts) > Boiler (9.4pts) > AC
(7.6pts)
Same day decisions: 83%
```

## Price-Conversion Dependency Dashboard




## Product vs Brand Analysis (Product Types × Brands)



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  DE DIETRICH  HITACHI  DOVRE  VIESSMANN  JOLLY MEC
SOL&MOI
main_equipment_category
AC                           1266                 2942    21              22
28             10            4      413      4          3          2        1
Boiler                        859                   28     4            1594
1540           1127          916        6      1        244          0        1
Heat Pump                    2126                 1378    39               0
31             37           13      508     10        329          1        7
Other                         587                    7    13             313
1             49          104       62      7         15          5      492
Stove                          22                    1  2662               0
1              0            0        4    639          0        548        0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  DE DIETRICH  HITACHI  DOVRE  VIESSMANN  JOLLY MEC
SOL&MOI
main_equipment_category
AC                           34.2                33.2  14.3            36.4
53.6           60.0          0.0     23.2   50.0       0.0       50.0      0.0
Boiler                       39.0                21.4  25.0            37.1
45.9           37.0         45.2     33.3  100.0      32.0        NaN      0.0
Heat Pump                    43.5                37.0  46.2             NaN
19.4           24.3         38.5     42.5   80.0      47.4      100.0     71.4
Other                        37.3                57.1  46.2            53.4
0.0           46.9         42.3     43.5   28.6      46.7       80.0     25.0
Stove                        27.3                 0.0  48.0             NaN
0.0            NaN          NaN     25.0   40.7       NaN       40.0      NaN

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
```

```
  Top 3 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. DOVRE: 80.0% (n=10)
    3. SOL&MOI: 71.4% (n=7)

Boiler:
  Best brand: DOVRE
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. DOVRE: 100.0% (n=1)
    2. FRISQUET: 45.9% (n=1,540)
    3. DE DIETRICH: 45.2% (n=916)

Stove:
  Best brand: MCZ
  Conversion: 48.0%
  Customers: 2,662
  Top 3 brands:
    1. MCZ: 48.0% (n=2,662)
    2. DOVRE: 40.7% (n=639)
    3. JOLLY MEC: 40.0% (n=548)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 60.0%
  Customers: 10
  Top 3 brands:
    1. SAUNIER DUVAL: 60.0% (n=10)
    2. FRISQUET: 53.6% (n=28)
    3. DOVRE: 50.0% (n=4)

Other:
  Best brand: JOLLY MEC
  Conversion: 80.0%
  Customers: 5
  Top 3 brands:
    1. JOLLY MEC: 80.0% (n=5)
    2. MITSUBISHI ELECTRIC: 57.1% (n=7)
    3. E.L.M. LEBLANC: 53.4% (n=313)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 43.5%
```

```
  Customers: 2,126
  All products:
    Heat Pump: 43.5% (n=2,126)
    Boiler: 39.0% (n=859)
    Other: 37.3% (n=587)
    AC: 34.2% (n=1,266)
    Stove: 27.3% (n=22)

MITSUBISHI ELECTRIC:
  Best product: Other
  Conversion: 57.1%
  Customers: 7
  All products:
    Other: 57.1% (n=7)
    Heat Pump: 37.0% (n=1,378)
    AC: 33.2% (n=2,942)
    Boiler: 21.4% (n=28)
    Stove: 0.0% (n=1)

MCZ:
  Best product: Stove
  Conversion: 48.0%
  Customers: 2,662
  All products:
    Stove: 48.0% (n=2,662)
    Other: 46.2% (n=13)
    Heat Pump: 46.2% (n=39)
    Boiler: 25.0% (n=4)
    AC: 14.3% (n=21)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 53.4%
  Customers: 313
  All products:
    Other: 53.4% (n=313)
    Boiler: 37.1% (n=1,594)
    AC: 36.4% (n=22)

FRISQUET:
  Best product: AC
  Conversion: 53.6%
  Customers: 28
  All products:
    AC: 53.6% (n=28)
    Boiler: 45.9% (n=1,540)
    Heat Pump: 19.4% (n=31)
    Other: 0.0% (n=1)
```

```
    Stove: 0.0% (n=1)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 60.0%
  Customers: 10
  All products:
    AC: 60.0% (n=10)
    Other: 46.9% (n=49)
    Boiler: 37.0% (n=1,127)
    Heat Pump: 24.3% (n=37)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.2%
  Customers: 916
  All products:
    Boiler: 45.2% (n=916)
    Other: 42.3% (n=104)
    Heat Pump: 38.5% (n=13)
    AC: 0.0% (n=4)

HITACHI:
  Best product: Other
  Conversion: 43.5%
  Customers: 62
  All products:
    Other: 43.5% (n=62)
    Heat Pump: 42.5% (n=508)
    Boiler: 33.3% (n=6)
    Stove: 25.0% (n=4)
    AC: 23.2% (n=413)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

Heat Pump:
  Volume leader: ATLANTIC (2,126 customers, 47.5% share)
  Conversion leader: JOLLY MEC (100.0%)

Boiler:
  Volume leader: E.L.M. LEBLANC (1,594 customers, 25.2% share)
  Conversion leader: DOVRE (100.0%)

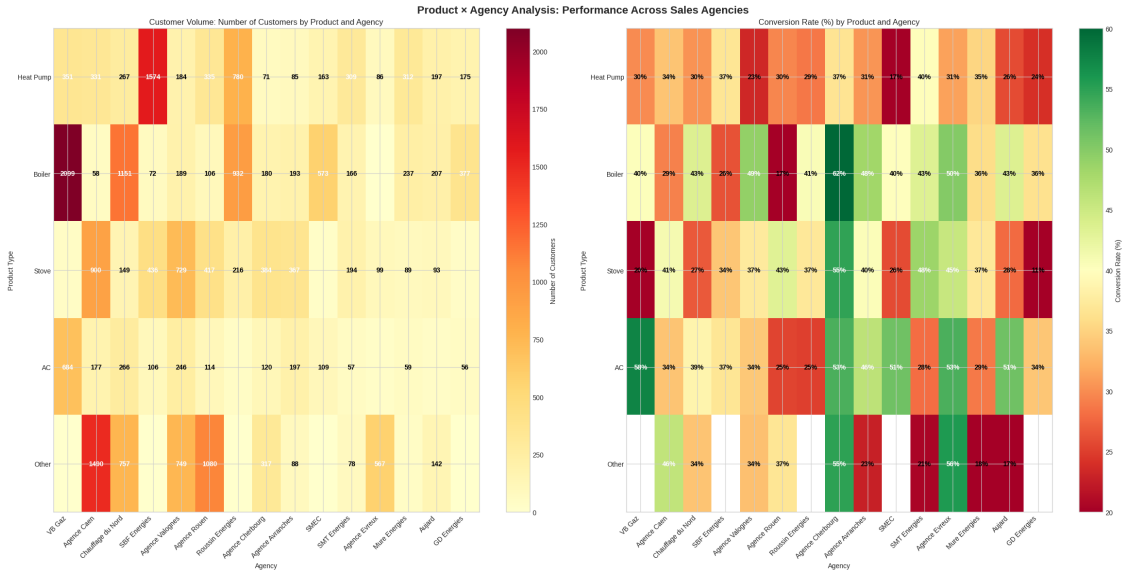
Stove:
```

```
  Volume leader: MCZ (2,662 customers, 68.7% share)
  Conversion leader: MCZ (48.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (2,942 customers, 62.4% share)
  Conversion leader: SAUNIER DUVAL (60.0%)

Other:
  Volume leader: ATLANTIC (587 customers, 35.5% share)
  Conversion leader: JOLLY MEC (80.0%)

Top 15 agencies by customer volume:
  VB Gaz: 3,175 customers, 42.5% conversion
  Agence Caen: 2,956 customers, 42.0% conversion
  Chauffage du Nord: 2,590 customers, 37.8% conversion
  SBF Energies: 2,188 customers, 35.9% conversion
  Agence Valognes: 2,097 customers, 35.5% conversion
  Agence Rouen: 2,052 customers, 35.2% conversion
  Roussin Energies: 1,972 customers, 35.1% conversion
  Agence Cherbourg: 1,072 customers, 54.7% conversion
  Agence Avranches: 930 customers, 40.6% conversion
  SMEC: 876 customers, 36.6% conversion
  SMT Energies: 804 customers, 39.9% conversion
  Agence Evreux: 790 customers, 51.6% conversion
  Mure Energies: 742 customers, 34.1% conversion
  Aujard: 676 customers, 31.1% conversion
  GD Energies: 644 customers, 31.4% conversion
```



```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              VB Gaz  Agence Caen  Chauffage du Nord  SBF Energies
Agence Valognes  Agence Rouen  Roussin Energies  Agence Cherbourg  Agence
Avranches  SMEC  SMT Energies  Agence Evreux  Mure Energies  Aujard  GD Energies
main_equipment_category
AC                         351          331                267          1574
184           335               780                71                85   163
309             86            312     197          175
Boiler                    2099           58               1151            72
189           106               932               180               193   573
166              8            237     207          377
Heat Pump                   41          900                149           436
729           417               216               384               367    31
194             99             89      93           36
Other                      684          177                266           106
246           114                44               120               197   109
57              30             59      37           56
Stove                       0         1490                757             0
749          1080                 0               317                88     0
78            567             45     142            0

================================================================================
  CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              VB Gaz  Agence Caen  Chauffage du Nord  SBF Energies
Agence Valognes  Agence Rouen  Roussin Energies  Agence Cherbourg  Agence
Avranches  SMEC  SMT Energies  Agence Evreux  Mure Energies  Aujard  GD Energies
main_equipment_category
AC                        29.9         33.8               30.3          36.8
23.4          29.6              28.7              36.6              30.6  17.2
39.8           31.4           35.3    25.9         24.0
Boiler                    40.2         29.3               43.4          26.4
49.2          17.0              40.6              62.2              48.2  40.0
43.4           50.0           35.9    43.5         36.3
Heat Pump                 19.5         41.0               26.8          33.9
37.4          43.2              37.0              54.7              40.3  25.8
48.5           45.5           37.1    28.0         11.1
Other                     57.6         33.9               38.7          36.8
34.1          25.4              25.0              53.3              46.2  51.4
28.1           53.3           28.8    51.4         33.9
Stove                      NaN         45.9               33.8           NaN
```

```
33.5          36.8                NaN              54.9              22.7   NaN
20.5           55.7           17.8    16.9          NaN

================================================================================
 BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------

Heat Pump:
  Best agency: Agence Cherbourg
  Conversion: 54.7%
  Customers: 384
  Top 3 agencies:
    1. Agence Cherbourg: 54.7% (n=384)
    2. SMT Energies: 48.5% (n=194)
    3. Agence Evreux: 45.5% (n=99)

Boiler:
  Best agency: Agence Cherbourg
  Conversion: 62.2%
  Customers: 180
  Top 3 agencies:
    1. Agence Cherbourg: 62.2% (n=180)
    2. Agence Evreux: 50.0% (n=8)
    3. Agence Valognes: 49.2% (n=189)

Stove:
  Best agency: Agence Evreux
  Conversion: 55.7%
  Customers: 567
  Top 3 agencies:
    1. Agence Evreux: 55.7% (n=567)
    2. Agence Cherbourg: 54.9% (n=317)
    3. Agence Caen: 45.9% (n=1,490)

AC:
  Best agency: SMT Energies
  Conversion: 39.8%
  Customers: 309
  Top 3 agencies:
    1. SMT Energies: 39.8% (n=309)
    2. SBF Energies: 36.8% (n=1,574)
    3. Agence Cherbourg: 36.6% (n=71)

Other:
  Best agency: VB Gaz
  Conversion: 57.6%
  Customers: 684
```

```
  Top 3 agencies:
    1. VB Gaz: 57.6% (n=684)
    2. Agence Evreux: 53.3% (n=30)
    3. Agence Cherbourg: 53.3% (n=120)

================================================================================
 BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

VB Gaz:
  Best product: Other
  Conversion: 57.6%
  Customers: 684
  All products:
    Other: 57.6% (n=684)
    Boiler: 40.2% (n=2,099)
    AC: 29.9% (n=351)
    Heat Pump: 19.5% (n=41)

Agence Caen:
  Best product: Stove
  Conversion: 45.9%
  Customers: 1,490
  All products:
    Stove: 45.9% (n=1,490)
    Heat Pump: 41.0% (n=900)
    Other: 33.9% (n=177)
    AC: 33.8% (n=331)
    Boiler: 29.3% (n=58)

Chauffage du Nord:
  Best product: Boiler
  Conversion: 43.4%
  Customers: 1,151
  All products:
    Boiler: 43.4% (n=1,151)
    Other: 38.7% (n=266)
    Stove: 33.8% (n=757)
    AC: 30.3% (n=267)
    Heat Pump: 26.8% (n=149)

SBF Energies:
  Best product: AC
  Conversion: 36.8%
  Customers: 1,574
  All products:
    AC: 36.8% (n=1,574)
```

```
    Other: 36.8% (n=106)
    Heat Pump: 33.9% (n=436)
    Boiler: 26.4% (n=72)

Agence Valognes:
  Best product: Boiler
  Conversion: 49.2%
  Customers: 189
  All products:
    Boiler: 49.2% (n=189)
    Heat Pump: 37.4% (n=729)
    Other: 34.1% (n=246)
    Stove: 33.5% (n=749)
    AC: 23.4% (n=184)

Agence Rouen:
  Best product: Heat Pump
  Conversion: 43.2%
  Customers: 417
  All products:
    Heat Pump: 43.2% (n=417)
    Stove: 36.8% (n=1,080)
    AC: 29.6% (n=335)
    Other: 25.4% (n=114)
    Boiler: 17.0% (n=106)

Roussin Energies:
  Best product: Boiler
  Conversion: 40.6%
  Customers: 932
  All products:
    Boiler: 40.6% (n=932)
    Heat Pump: 37.0% (n=216)
    AC: 28.7% (n=780)
    Other: 25.0% (n=44)

Agence Cherbourg:
  Best product: Boiler
  Conversion: 62.2%
  Customers: 180
  All products:
    Boiler: 62.2% (n=180)
    Stove: 54.9% (n=317)
    Heat Pump: 54.7% (n=384)
    Other: 53.3% (n=120)
    AC: 36.6% (n=71)

Agence Avranches:
```

```
  Best product: Boiler
  Conversion: 48.2%
  Customers: 193
  All products:
    Boiler: 48.2% (n=193)
    Other: 46.2% (n=197)
    Heat Pump: 40.3% (n=367)
    AC: 30.6% (n=85)
    Stove: 22.7% (n=88)

SMEC:
  Best product: Other
  Conversion: 51.4%
  Customers: 109
  All products:
    Other: 51.4% (n=109)
    Boiler: 40.0% (n=573)
    Heat Pump: 25.8% (n=31)
    AC: 17.2% (n=163)

================================================================================
 AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                   conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
VB Gaz                        42.5            3175            2.1
1.1   4816.826
Agence Caen                   42.0            2956           32.2
40.6   8911.067
Chauffage du Nord             37.8            2590            7.0
35.5   6273.223
SBF Energies                  35.9            2188           24.8
15.2   9357.330
Agence Valognes               35.5            2097           38.5
44.7  10062.613
Agence Rouen                  35.2            2052           22.8
45.2   8705.339
Roussin Energies              35.1            1972           13.2
10.6   7387.763
Agence Cherbourg              54.7            1072           43.6
49.1   9332.296
Agence Avranches              40.6             930           45.1
38.1  10617.664
SMEC                          36.6             876            4.8
6.6   6215.490
```

```
SMT Energies                    39.9            804           29.7
40.7   7324.780
Agence Evreux                   51.6            790           13.2
47.0   7378.743
Mure Energies                   34.1            742           15.0
14.7   6710.125
Aujard                          31.1            676           16.0
11.5   8274.465
GD Energies                     31.4            644            9.0
7.6   5589.599

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Heat Pump:
  Conversion leader: Agence Cherbourg (54.7%, n=384)

Boiler:
  Conversion leader: Agence Cherbourg (62.2%, n=180)

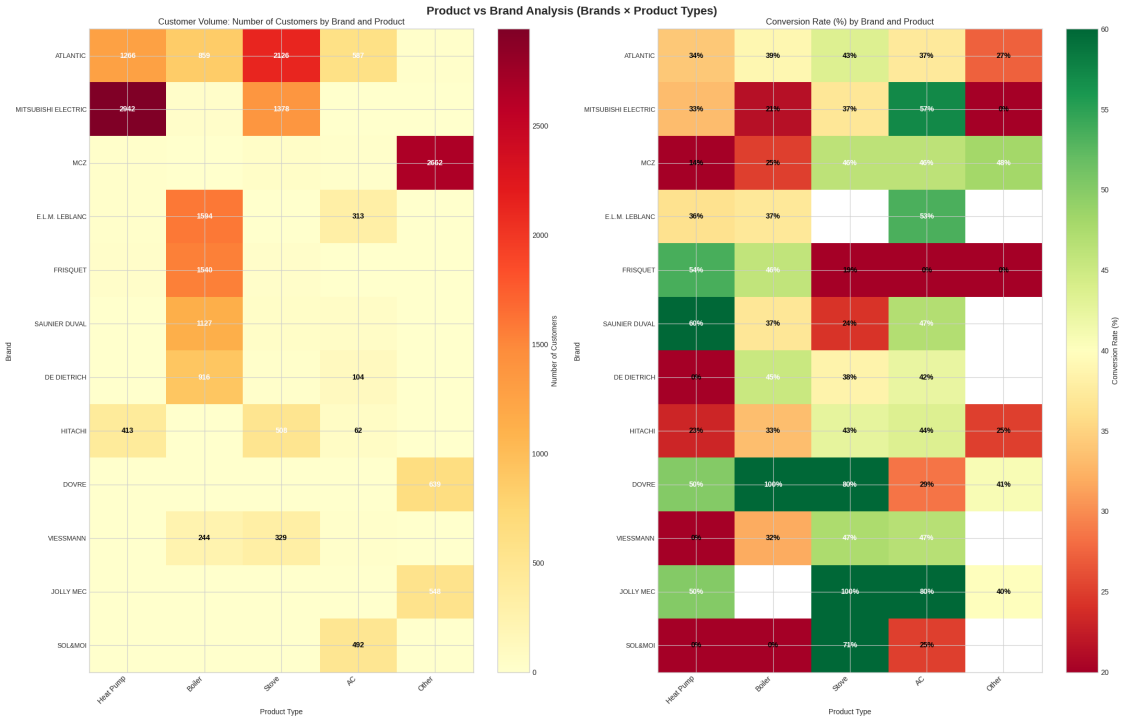
Stove:
  Conversion leader: Agence Evreux (55.7%, n=567)

AC:
  Conversion leader: SMT Energies (39.8%, n=309)

Other:
  Conversion leader: VB Gaz (57.6%, n=684)

================================================================================
 HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  Agence Cherbourg: 54.7% (n=384)
  SMT Energies: 48.5% (n=194)
  Agence Evreux: 45.5% (n=99)
  Agence Rouen: 43.2% (n=417)
  Agence Caen: 41.0% (n=900)
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                 1266     859       2126    587     22
MITSUBISHI ELECTRIC      2942      28       1378      7      1
MCZ                        21       4         39     13   2662
E.L.M. LEBLANC             22    1594          0    313      0
FRISQUET                   28    1540         31      1      1
SAUNIER DUVAL              10    1127         37     49      0
DE DIETRICH                 4     916         13    104      0
HITACHI                   413       6        508     62      4
DOVRE                       4       1         10      7    639
VIESSMANN                   3     244        329     15      0
JOLLY MEC                   2       0          1      5    548
SOL&MOI                     1       1          7    492      0

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                 34.2    39.0       43.5   37.3   27.3
MITSUBISHI ELECTRIC      33.2    21.4       37.0   57.1    0.0
MCZ                      14.3    25.0       46.2   46.2   48.0
E.L.M. LEBLANC           36.4    37.1        NaN   53.4    NaN
FRISQUET                 53.6    45.9       19.4    0.0    0.0
SAUNIER DUVAL            60.0    37.0       24.3   46.9    NaN
DE DIETRICH               0.0    45.2       38.5   42.3    NaN
HITACHI                  23.2    33.3       42.5   43.5   25.0
DOVRE                    50.0   100.0       80.0   28.6   40.7
VIESSMANN                 0.0    32.0       47.4   46.7    NaN
JOLLY MEC                50.0     NaN      100.0   80.0   40.0
SOL&MOI                   0.0     0.0       71.4   25.0    NaN

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Heat Pump
  Conversion: 43.5%
  Customers: 2,126
  Top 3 products:
    1. Heat Pump: 43.5% (n=2,126)
    2. Boiler: 39.0% (n=859)
    3. Other: 37.3% (n=587)

MITSUBISHI ELECTRIC:
  Best product: Other
  Conversion: 57.1%
  Customers: 7
  Top 3 products:
    1. Other: 57.1% (n=7)
    2. Heat Pump: 37.0% (n=1,378)
    3. AC: 33.2% (n=2,942)

MCZ:
  Best product: Stove
  Conversion: 48.0%
  Customers: 2,662
  Top 3 products:
    1. Stove: 48.0% (n=2,662)
    2. Other: 46.2% (n=13)
    3. Heat Pump: 46.2% (n=39)

E.L.M. LEBLANC:
```

```
  Best product: Other
  Conversion: 53.4%
  Customers: 313
  Top 3 products:
    1. Other: 53.4% (n=313)
    2. Boiler: 37.1% (n=1,594)
    3. AC: 36.4% (n=22)

FRISQUET:
  Best product: AC
  Conversion: 53.6%
  Customers: 28
  Top 3 products:
    1. AC: 53.6% (n=28)
    2. Boiler: 45.9% (n=1,540)
    3. Heat Pump: 19.4% (n=31)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 60.0%
  Customers: 10
  Top 3 products:
    1. AC: 60.0% (n=10)
    2. Other: 46.9% (n=49)
    3. Boiler: 37.0% (n=1,127)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.2%
  Customers: 916
  Top 3 products:
    1. Boiler: 45.2% (n=916)
    2. Other: 42.3% (n=104)
    3. Heat Pump: 38.5% (n=13)

HITACHI:
  Best product: Other
  Conversion: 43.5%
  Customers: 62
  Top 3 products:
    1. Other: 43.5% (n=62)
    2. Heat Pump: 42.5% (n=508)
    3. Boiler: 33.3% (n=6)

DOVRE:
  Best product: Boiler
  Conversion: 100.0%
  Customers: 1
```

```
  Top 3 products:
    1. Boiler: 100.0% (n=1)
    2. Heat Pump: 80.0% (n=10)
    3. AC: 50.0% (n=4)

VIESSMANN:
  Best product: Heat Pump
  Conversion: 47.4%
  Customers: 329
  Top 3 products:
    1. Heat Pump: 47.4% (n=329)
    2. Other: 46.7% (n=15)
    3. Boiler: 32.0% (n=244)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: JOLLY MEC
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. JOLLY MEC: 100.0% (n=1)
    2. DOVRE: 80.0% (n=10)
    3. SOL&MOI: 71.4% (n=7)
    4. VIESSMANN: 47.4% (n=329)
    5. MCZ: 46.2% (n=39)

Boiler:
  Best brand: DOVRE
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. DOVRE: 100.0% (n=1)
    2. FRISQUET: 45.9% (n=1,540)
    3. DE DIETRICH: 45.2% (n=916)
    4. ATLANTIC: 39.0% (n=859)
    5. E.L.M. LEBLANC: 37.1% (n=1,594)

Stove:
  Best brand: MCZ
  Conversion: 48.0%
  Customers: 2,662
  Top 5 brands:
    1. MCZ: 48.0% (n=2,662)
    2. DOVRE: 40.7% (n=639)
    3. JOLLY MEC: 40.0% (n=548)
```

```
    4. ATLANTIC: 27.3% (n=22)
    5. HITACHI: 25.0% (n=4)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 60.0%
  Customers: 10
  Top 5 brands:
    1. SAUNIER DUVAL: 60.0% (n=10)
    2. FRISQUET: 53.6% (n=28)
    3. DOVRE: 50.0% (n=4)
    4. JOLLY MEC: 50.0% (n=2)
    5. E.L.M. LEBLANC: 36.4% (n=22)

Other:
  Best brand: JOLLY MEC
  Conversion: 80.0%
  Customers: 5
  Top 5 brands:
    1. JOLLY MEC: 80.0% (n=5)
    2. MITSUBISHI ELECTRIC: 57.1% (n=7)
    3. E.L.M. LEBLANC: 53.4% (n=313)
    4. SAUNIER DUVAL: 46.9% (n=49)
    5. VIESSMANN: 46.7% (n=15)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Heat Pump: 43.5% (n=2,126)
  Boiler: 39.0% (n=859)
  Other: 37.3% (n=587)

MITSUBISHI ELECTRIC:
  Other: 57.1% (n=7)
  Heat Pump: 37.0% (n=1,378)
  AC: 33.2% (n=2,942)

MCZ:
  Stove: 48.0% (n=2,662)
  Other: 46.2% (n=13)
  Heat Pump: 46.2% (n=39)

E.L.M. LEBLANC:
  Other: 53.4% (n=313)
```

```
  Boiler: 37.1% (n=1,594)
  AC: 36.4% (n=22)

FRISQUET:
  AC: 53.6% (n=28)
  Boiler: 45.9% (n=1,540)
  Heat Pump: 19.4% (n=31)

SAUNIER DUVAL:
  AC: 60.0% (n=10)
  Other: 46.9% (n=49)
  Boiler: 37.0% (n=1,127)

DE DIETRICH:
  Boiler: 45.2% (n=916)
  Other: 42.3% (n=104)
  Heat Pump: 38.5% (n=13)

HITACHI:
  Other: 43.5% (n=62)
  Heat Pump: 42.5% (n=508)
  Boiler: 33.3% (n=6)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (2,126 customers, 47.5% share)
  Conversion leader: JOLLY MEC (100.0%)

Boiler:
  Volume leader: E.L.M. LEBLANC (1,594 customers, 25.2% share)
  Conversion leader: DOVRE (100.0%)

Stove:
  Volume leader: MCZ (2,662 customers, 68.7% share)
  Conversion leader: MCZ (48.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (2,942 customers, 62.4% share)
  Conversion leader: SAUNIER DUVAL (60.0%)

Other:
  Volume leader: ATLANTIC (587 customers, 35.5% share)
  Conversion leader: JOLLY MEC (80.0%)
```

```
(main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
 main_brand
 ATLANTIC                 1266     859       2126    587     22
 MITSUBISHI ELECTRIC      2942      28       1378      7      1
 MCZ                        21       4         39     13   2662
 E.L.M. LEBLANC             22    1594          0    313      0
 FRISQUET                   28    1540         31      1      1
 SAUNIER DUVAL              10    1127         37     49      0
 DE DIETRICH                 4     916         13    104      0
 HITACHI                   413       6        508     62      4
 DOVRE                       4       1         10      7    639
 VIESSMANN                   3     244        329     15      0
 JOLLY MEC                   2       0          1      5    548
 SOL&MOI                     1       1          7    492      0,
 main_equipment_category         AC      Boiler   Heat Pump      Other  \
 main_brand
 ATLANTIC                 34.202212   38.998836   43.461900  37.308348
 MITSUBISHI ELECTRIC      33.208702   21.428571   37.010160  57.142857
 MCZ                      14.285714   25.000000   46.153846  46.153846
 E.L.M. LEBLANC           36.363636   37.139272         NaN  53.354633
 FRISQUET                 53.571429   45.909091   19.354839   0.000000
 SAUNIER DUVAL            60.000000   37.000887   24.324324  46.938776
 DE DIETRICH               0.000000   45.196507   38.461538  42.307692
 HITACHI                  23.244552   33.333333   42.519685  43.548387
 DOVRE                    50.000000  100.000000   80.000000  28.571429
 VIESSMANN                 0.000000   31.967213   47.416413  46.666667
 JOLLY MEC                50.000000         NaN  100.000000  80.000000
 SOL&MOI                   0.000000    0.000000   71.428571  25.000000

 main_equipment_category      Stove
 main_brand
 ATLANTIC                 27.272727
 MITSUBISHI ELECTRIC       0.000000
 MCZ                      48.046582
 E.L.M. LEBLANC                 NaN
 FRISQUET                  0.000000
 SAUNIER DUVAL                  NaN
 DE DIETRICH                    NaN
 HITACHI                  25.000000
 DOVRE                    40.688576
 VIESSMANN                      NaN
 JOLLY MEC                39.963504
 SOL&MOI                        NaN  )
```

```
import ipynbname
import subprocess
import os
```

```python
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
```

```
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
 Notebook: _eda_price.ipynb
 PDF path: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price.pdf
 Removing existing PDF: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price.pdf
 Converting to PDF…
 PDF saved: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price.pdf
```